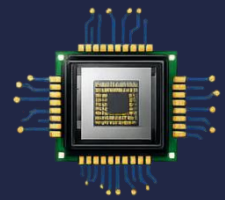


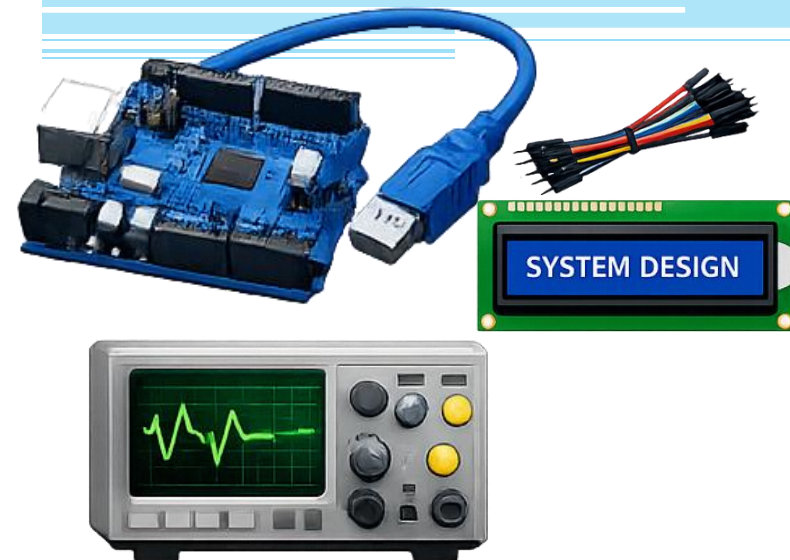
CS-301

Microprocessor- Based System Design



Lecture 7: Interfacing Microprocessors with Basic IO Devices (LCDs)

Course Teacher: Syeda Ramish Fatima



Intefacing of ATmega162 with LCD Controller



16x2 LCD Module



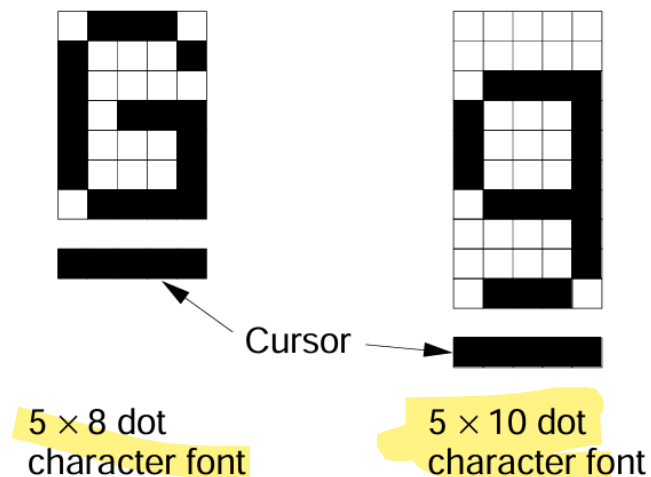
16x2 LCD Module Pinout

Brief Description on LCD modules

- LCD modules are very commonly used in most embedded projects,
 - reason being its cheap price, availability and programmer friendly.
- Most of us would have come across these displays in our day to day life,
 - PC's or calculators, front panels (user interface) of embedded applications

Brief Description on LCD modules

- **16×2 LCD** is named so because; it has 16 Columns and 2 Rows.
- There are a lot of combinations available like, 8×1, 8×2, 10×2, 16×1, etc. but the most used one is the 16×2 LCD.
- So, it will have (16×2=32) 32 characters in total and each character will be made of 5×8 Pixel Dots.
- A Single character with all its Pixels is shown in the below picture.



Cursor display example

Brief Description on LCD modules

- Since each character has ($5 \times 8 = 40$) 40 Pixels, for 32 Characters we will have (32×40) 1280 Pixels.
- LCD should be instructed about the Position of the Pixels.
- It will be a hectic task to handle everything with the help of MCU,
- An **Interface IC like HD44780** is used, which is mounted on the backside of the LCD Module itself.
- The function of this IC is to get the **Commands and Data** from the MCU and process them to display meaningful information onto our LCD Screen. Microcontroller Unit. (MCU)
- You can learn how to interface an LCD.
- If you are an advanced programmer and would like to create your own library for interfacing your Microcontroller with this LCD module then you have to understand the HD44780 IC's working and commands which can be found in its datasheet [HD44780.pdf](#) .

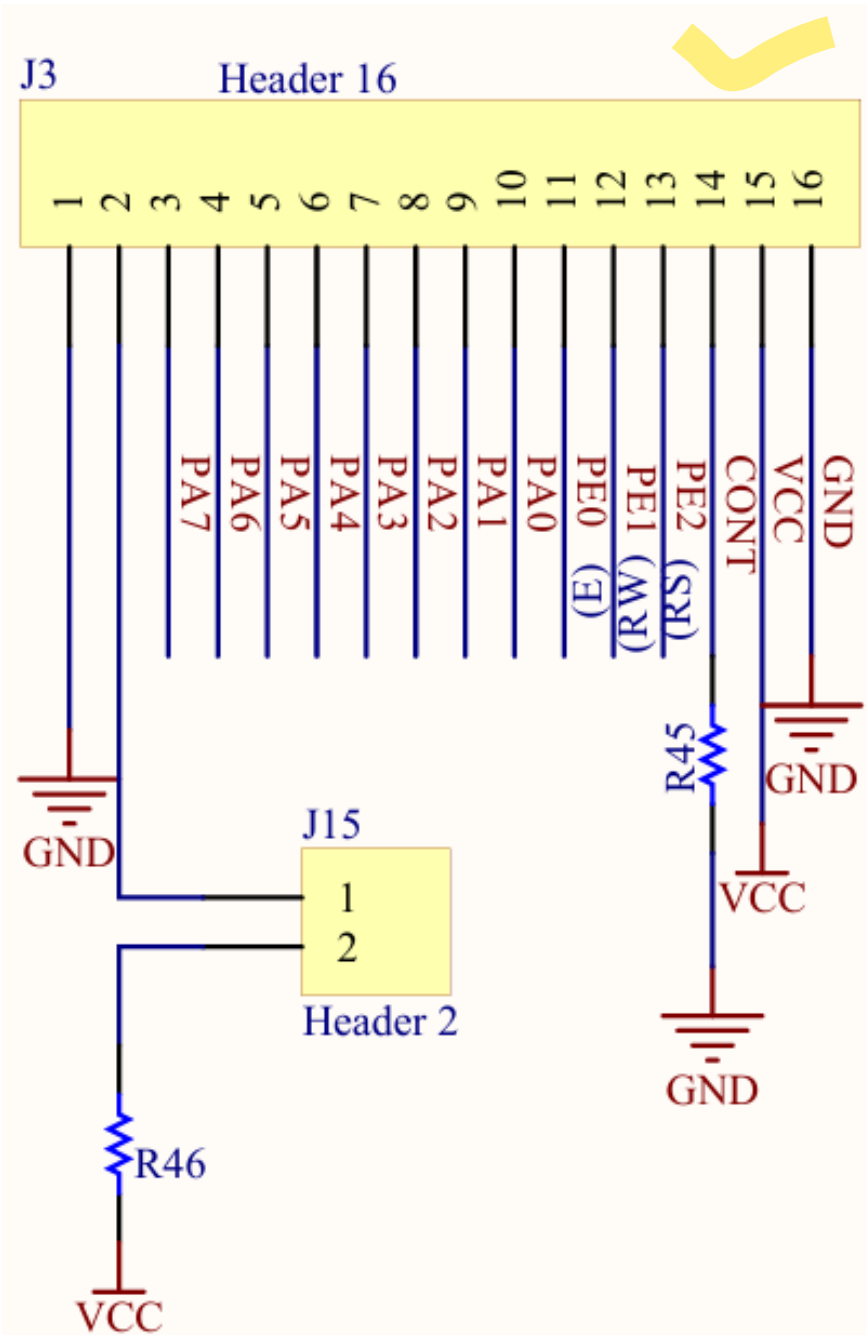
Interfacing LCD module with ATmega162(PCB Schematic)

pa0 -pa7 : mcu pins to header 16: data commnad lines

CONT (R45)

Pull-down resistor to GND via R45 resistor

pull down resistor : active low when no inpt given





- *Further Reading: Go through the datasheet as many times as needed!*
- ***You can avoid reality, for a while. But you can't avoid the consequences of reality!** - C.W. Livingston*